A Relative Schur Classification of Finite-Order Weak–Strong Differentiability Principles

Automatic mathematics run `fa_banach_001`

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Status. Candidate strong partial result, likely valid, for human review. For Banach targets it completely classifies the finite-order weak–strong principle for each boundedness-determining pair (E, G). It answers the full-dual Banach part of Question 6.9(i) affirmatively (already covered by Bachir–Lancien's 2003 theorem) and extends the classification to arbitrary $G \subset E'$ that determine boundedness. It does not classify all locally convex targets or spaces that work uniformly for every admissible G.

1 Source question

The source is Karsten Kruse, *Extension of vector-valued functions and weak-strong principles for differentiable functions of finite order*, arXiv:1910.01952 [1]. Theorem 6.6 proves the full finite-order principle when E is semi-Montel and $G \subset E'$ determines boundedness. Question 6.9(i), on printed page 21, asks whether there are other target spaces and whether the positive C^0 result for spaces whose bounded sets are precompact extends to positive finite order. The complete source question is reproduced below.

> everywhere because E is a generalised Gelfand space, implying that $(\partial^{k-1})^E f$ is differentiable on $[a,b]$ almost everywhere. Since the open set $\Omega \subset \mathbb{R}$ can be written as a countable union of disjoint open intervals I_n, $n \in \mathbb{N}$, and each I_n is a countable union of closed bounded intervals $[a_m, b_m]$, $m \in \mathbb{N}$, our statement follows from the fact that the countable union of null sets is a null set. □
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> To the best of our knowledge there are still some open problems for continuously partially differentiable functions of finite order.
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> 6.9. **Question.** (i) Are there other spaces than semi-Montel spaces E for which the 'full' $\mathcal{C}^k$-weak-strong principle Theorem 6.6 with $k < \infty$ is true? For instance, if $k = 0$, then it is still true if E is an lcHs such that every bounded set is already precompact in E by [6, 2.10 Lemma, p. 140]. Does this hold for $0 < k < \infty$ as well?
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> (ii) Does the 'almost' $\mathcal{C}^k$-weak-strong principle Corollary 6.8 also hold for $d > 1$?
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> (iii) For every $\varepsilon > 0$ does there exist a function $g \in \mathcal{C}^k(\mathbb{R}, E)$ such that $\lambda(\{x \in \Omega \mid f(x) \neq g(x)\}) < \varepsilon$ in Corollary 6.8 where λ is the one-dimensional Lebesgue measure. In the case that $E = \mathbb{R}^n$ this is true by [23, Theorem 3.1.15, p. 227].
>
> (iv) Is there a 'Radon–Nikodým type' characterisation of generalised Gelfand spaces as in the Banach case?
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> 7. Vector-valued Blaschke theorems
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> In this section we prove several convergence theorems for Banach-valued func-

We solve a precise Banach-space slice of the first sentence. The result also shows that, once the theorem permits a proper test set $G \subset E'$, the answer depends on the pair (E, G) and not only on E.

2 Definitions and statement

Let E be a real or complex Banach space and let $G \subset E'$ determine boundedness, meaning that a set $B \subset E$ is norm bounded whenever

$$\sup_{x \in B} |g(x)| < \infty \qquad \text{for every } g \in G. \tag{1}$$

This already forces G to separate points: if every $g \in G$ annihilates x, then $\{nx : n \in \mathbb{N}\}$ is G-bounded, so (1) implies $x = 0$.

We say that (E, G) has the *relative sequential Schur property* if

$$(\mathrm{S}_G) \qquad \left.\begin{array}{r} (x_n) \subset E \text{ is norm bounded,} \\ g(x_n) \longrightarrow 0 \text{ for every } g \in G \end{array}\right\} \Longrightarrow \|x_n\| \longrightarrow 0. \tag{2}$$

Because G determines boundedness, every G-null sequence is automatically norm bounded. Thus the boundedness line in (2) is logically redundant here, but records the standard relative Schur formulation.

Theorem 1 (Banach-pair classification)**.** *Let E be a Banach space, let $G \subset E'$ determine boundedness, and fix $k \in \mathbb{N}$. The following are equivalent.*

(i) The pair (E, G) has property (S_G).

(ii) For every real Banach space Y, every open $U \subset Y$, and every map $f : U \to E$, the conditions

$$g \circ f \in C^k(U, \mathbb{K}) \qquad (g \in G)$$

imply $f \in C^k(U, E)$ in the Fréchet sense.

(iii) The implication in (ii) holds just for $Y = \mathbb{R}$ and open intervals $U \subset \mathbb{R}$.

In particular, the same relative Schur condition characterizes the principle for every positive finite order, and one-dimensional curves already detect failure.

The theorem is stronger in its domain than the source question, which only requires $U \subset \mathbb{R}^d$.

3 Proof intuition

There are two short mechanisms. First, G-scalar smoothness makes all the relevant differences, difference quotients, Taylor remainders, and derivative differences converge coordinatewise under G. Condition (1) makes each such sequence norm bounded, while (S_G) upgrades its G-convergence to norm convergence. This constructs the derivatives and proves their continuity.

Conversely, if (S_G) fails, take a bounded G-null sequence (x_n) that stays away from zero. Put increasingly narrow smooth bumps with values in the directions x_n at points $t_n \downarrow 0$. Scale the nth bump by the kth power of its width. Its first $k-1$ derivatives then disappear at the accumulation point. The kth derivative still takes the value x_n at the bump centers, so it is not norm continuous; after applying any fixed $g \in G$, those values become $g(x_n) \to 0$, so every scalarization is nevertheless C^k.

4 Proof of the classification

Proof of Theorem 1. The implication (ii)⇒(iii) is immediate. We first prove (i)⇒(ii), and then build a counterexample proving (iii)⇒(i).

Step 1: the case $k = 1$. Suppose (S_G) holds and $g \circ f \in C^1(U)$ for every $g \in G$. If $y_n \to y$ in U, then

$$g(f(y_n) - f(y)) \longrightarrow 0 \qquad (g \in G).$$

This difference sequence is norm bounded by (1), and hence converges to zero in norm by (S_G). Since U and E are metrizable, f is continuous.

Fix $y \in U$ and $h \in Y$. For nonzero t sufficiently close to zero put

$$q_t(h) = \frac{f(y + th) - f(y)}{t}.$$

If these quotients were not norm Cauchy as $t \to 0$, there would be $s_n, t_n \to 0$ and $\varepsilon > 0$ such that $\|q_{s_n}(h) - q_{t_n}(h)\| \geq \varepsilon$. But for every $g \in G$,

$$g(q_{s_n}(h) - q_{t_n}(h)) \longrightarrow 0$$

because $g \circ f$ is differentiable at y. Boundedness determination and (S_G) give a contradiction. Completeness of E therefore gives a limit

$$A_y h := \lim_{t \to 0} q_t(h) \in E, \qquad g(A_y h) = D(g \circ f)(y)h. \tag{3}$$

Since G separates points, (3) makes A_y linear. Moreover,

$$\sup_{\|h\| \leq 1} |g(A_y h)| = \|D(g \circ f)(y)\| < \infty \qquad (g \in G).$$

Thus A_y maps the unit ball of Y to a G-bounded set, and (1) shows that $A_y \in \mathcal{L}(Y, E)$.

We next show that A_y is the Fréchet derivative. If not, there are $h_n \to 0$ with $h_n \neq 0$ and $\varepsilon > 0$ such that

$$z_n := \frac{f(y + h_n) - f(y) - A_y h_n}{\|h_n\|}, \qquad \|z_n\| \geq \varepsilon. \tag{4}$$

Equation (3) and scalar Fréchet differentiability give $g(z_n) \to 0$ for every $g \in G$. Again (1) and (S_G) contradict (4). Hence $Df(y) = A_y$.

Finally Df is operator-norm continuous. Otherwise, for some $y_n \to y$ we could choose $\|h_n\| = 1$ and $\varepsilon > 0$ with

$$\|(Df(y_n) - Df(y))h_n\| \geq \varepsilon.$$

For each $g \in G$ the left-hand vectors are G-null, since

$$|g((Df(y_n) - Df(y))h_n)| \leq \|D(g \circ f)(y_n) - D(g \circ f)(y)\| \longrightarrow 0.$$

This contradicts (S_G). We have proved $f \in C^1(U, E)$.

Step 2: induction on the order. Assume the assertion has been proved through order $m - 1$, where $m \geq 2$, and suppose every $g \circ f$ is C^m. By induction $f \in C^{m-1}$. Write

$$F(y) = D^{m-1} f(y) \in \mathcal{L}^{m-1}(Y, E).$$

For fixed y and $v_1, \ldots, v_m \in Y$, apply the quotient argument from Step 1 to

$$\frac{[F(y+tv_1) - F(y)](v_2, \ldots, v_m)}{t}.$$

It converges in E; call its limit $B_y(v_1, \ldots, v_m)$. Scalarization gives

$$g(B_y(v_1, \ldots, v_m)) = D^m(g \circ f)(y)(v_1, \ldots, v_m). \tag{5}$$

Separation by G makes B_y m-linear. For each g, (5) is bounded on the product of the unit balls by $\|D^m(g \circ f)(y)\|$. Thus (1) makes the whole image norm bounded, and $B_y \in \mathcal{L}^m(Y, E)$.

The map F is Fréchet differentiable at y with derivative $h \mapsto B_y(h, \cdot)$. Indeed, failure in operator norm would give $h_n \to 0$ and unit vectors $v_{2,n}, \ldots, v_{m,n}$ such that the E-valued normalized remainders

$$\frac{[F(y+h_n) - F(y) - B_y(h_n, \cdot)](v_{2,n}, \ldots, v_{m,n})}{\|h_n\|} \tag{6}$$

stay away from zero. Applying g bounds (6) by the operator norm of the normalized Fréchet remainder of $D^{m-1}(g \circ f)$, which tends to zero. Property (S_G) again contradicts the assumed lower bound.

The same unit-vector argument proves continuity of $y \mapsto B_y$ in $\mathcal{L}^m(Y, E)$: if it failed along $y_n \to y$, evaluate $B_{y_n} - B_y$ on unit m-tuples witnessing the failure and use continuity of every scalar $D^m(g \circ f)$. Therefore $F \in C^1$ and $f \in C^m$. Induction proves (i)⇒(ii) for the fixed k.

Step 3: a smooth-bump converse. Suppose (S_G) fails. After passing to a subsequence, there are a bounded sequence $(x_n) \subset E$ and $\delta > 0$ such that

$$g(x_n) \longrightarrow 0 \quad (g \in G), \qquad \|x_n\| \geq \delta \quad (n \in \mathbb{N}). \tag{7}$$

Choose $\phi \in C_c^\infty((-1,1))$ with $\phi^{(k)}(0) = 1$. Let

$$t_n = 2^{-n}, \qquad w_n = 2^{-3n}, \qquad I_n = (t_n - w_n, t_n + w_n).$$

The intervals I_n are pairwise disjoint and accumulate only at zero. Set

$$f(t) = \begin{cases} w_n^k \, \phi\left(\dfrac{t - t_n}{w_n}\right) x_n, & t \in I_n, \\ 0, & t \notin \bigcup_n I_n. \end{cases} \tag{8}$$

Compact support of ϕ makes (8) smooth across every endpoint of I_n. For $t \in I_n$ and $0 \leq j \leq k$,

$$f^{(j)}(t) = w_n^{k-j} \phi^{(j)}\left(\frac{t - t_n}{w_n}\right) x_n. \tag{9}$$

For $0 \leq j \leq k-1$, boundedness of (x_n) and (9) give

$$\frac{\|f^{(j)}(t)\|}{|t|} \leq C_j \frac{w_n^{k-j}}{t_n - w_n} \leq 2C_j \frac{w_n}{t_n} = 2C_j 2^{-2n} \longrightarrow 0. \tag{10}$$

Starting with $j = 0$ and proceeding successively, (10) proves that all derivatives through order k exist at zero and equal zero; (9) also gives continuity there through order $k-1$.

For any fixed $g \in G$, the scalar kth derivative on I_n is

$$(g \circ f)^{(k)}(t) = \phi^{(k)}\left(\frac{t - t_n}{w_n}\right) g(x_n).$$

Its supremum on I_n tends to zero by (7). Thus $g \circ f \in C^k(\mathbb{R})$ for every $g \in G$. On the other hand,

$$f^{(k)}(t_n) = x_n, \qquad f^{(k)}(0) = 0,$$

so (7) says that $f^{(k)}$ is not norm continuous at zero. Consequently $f \notin C^k(\mathbb{R}, E)$, contradicting (iii). Hence (iii) implies (S_G) and the proof is complete. □

5 Consequences and sharpness

Corollary 1 (Full dual)**.** *For a Banach space E and any fixed $k \geq 1$, every weakly C^k map from an open subset of any Banach space into E is C^k if and only if E has the Schur property.*

For $G = E'$, condition (S_G) is exactly the ordinary Schur property. The positive implication, including arbitrary Banach domains and all finite orders, is Corollaries 2.2–2.3 of Bachir–Lancien [2]; their Proposition 2.4 also gives a differentiability-level converse. Thus Corollary 1 is not claimed as new.

Taking $E = \ell_1$ and $G = E' = \ell_\infty$ gives an infinite-dimensional affirmative example for the first sentence of Question 6.9(i) in the full-dual Banach setting. It is not semi-Montel: an infinite-dimensional Banach space has non-precompact closed unit ball.

The dependence on G is sharp even for that same E. Regard $c_0 \subset \ell_\infty = (\ell_1)'$. The set c_0 determines boundedness of ℓ_1: if $B \subset \ell_1$ is pointwise bounded on c_0, view its members as continuous linear functionals on the Banach space c_0 and apply uniform boundedness. But the unit vectors (e_n) satisfy

$$\langle e_n, a \rangle = a_n \longrightarrow 0 \qquad (a \in c_0), \qquad \|e_n\|_1 = 1.$$

Hence (ℓ_1, c_0) fails (S_G), and (8) gives an explicit weak–strong counterexample for every finite $k \geq 1$. Therefore no classification that mentions only the Banach target E can describe all admissible proper test sets G.

6 Literature and novelty bounds

Bachir–Lancien's 2003 paper [2] proves the full-dual Schur case and is the essential closest result. The argument above deliberately credits that mechanism; its candidate advance is the exact equivalence for an arbitrary boundedness-determining pair (E, G), the reduction to curves, and the sharp contrast between (ℓ_1, ℓ_∞) and (ℓ_1, c_0) in the framework of Kruse's question.

On 9 August 2026, bounded searches of the run indexes, the local full-source arXiv corpus, and official-arXiv/web results used the source title and the terms "weakly C^k," "weak–strong principle," "Schur property," "τ-Schur," and "$\sigma(E, G)$-Schur." They recovered the 2003 full-dual theorem and literature on relative Schur topologies, but no source stating the differentiability classification for boundedness-determining G. This is a limited novelty audit, not publication-level certification.

7 Scope and human review

Theorem 1 is a complete Banach-pair result and a substantial partial answer to Question 6.9(i). It does not settle the locally convex precompact-bounded-set question, Questions 6.9(ii)–(iv), or the possible classification of Banach spaces for which the principle holds simultaneously for every G that determines boundedness.

Human review should focus on the higher-order operator-norm induction, the interpretation of the source question when G varies, and whether the relative extension has appeared under τ-Schur terminology. The bump construction and the c_0 boundedness-determination example are also worth checking independently.

References

[1] K. Kruse, *Extension of vector-valued functions and weak-strong principles for differentiable functions of finite order*, arXiv:1910.01952. `https://arxiv.org/abs/1910.01952`

[2] M. Bachir and G. Lancien, *On the Composition of Differentiable Functions*, Canadian Mathematical Bulletin **46** (2003), 481–494. `https://doi.org/10.4153/CMB-2003-047-2`